

Module 11: Automation and Programmability (Detailed Notes)

1. How Automation Impacts Network Management

Automation in network management reduces the need for manual configuration of devices. It minimizes human errors, improves consistency, and speeds up deployment of network services. Automation allows administrators to manage large-scale networks efficiently through scripts and centralized tools. It also enables proactive monitoring, faster troubleshooting, and better security enforcement.

2. Traditional Network vs Controller-Based Networking

Traditional Networking: - Each device is configured individually using CLI. - Management is decentralized. - Changes are time-consuming and error-prone. Controller-Based Networking: - Uses a centralized controller to manage the entire network. - Devices receive configurations automatically. - Provides automation, scalability, and consistency. - Easier to enforce policies across the network.

3. Virtualization

Virtualization is the process of creating virtual instances of physical resources such as servers, storage, and networks. It allows multiple virtual machines to run on a single physical device using a hypervisor. Benefits include better resource utilization, cost reduction, flexibility, scalability, and easier disaster recovery.

4. Characteristics of REST-based APIs

REST (Representational State Transfer) APIs are used for communication between systems. Key characteristics: - Stateless: Each request contains all necessary information. - Uses HTTP methods: GET, POST, PUT, DELETE. - Data formats: JSON and XML. - Scalable and lightweight. - Easy integration with automation tools and scripts.

5. Methods of Automation

Automation can be achieved using different methods: - Scripting: Using languages like Python to automate tasks. - Configuration Management Tools: Ansible, Puppet, Chef for managing configurations. - Orchestration Tools: Coordinate multiple automated tasks. - APIs: Interact with devices and controllers programmatically.

6. Software-Defined Networking (SDN)

SDN is an approach that separates the control plane from the data plane. - Control Plane: Makes decisions about where traffic is sent. - Data Plane: Forwards traffic. SDN provides centralized

control, programmability, and dynamic network management, making networks more flexible and efficient.

7. Cisco DNA Center

Cisco DNA Center is a centralized network management and automation platform. Features: - Network automation and provisioning. - Policy-based management. - Network assurance and analytics. - Security integration. It simplifies network operations and improves visibility and performance.

8. SD-Access and SD-WAN

SD-Access: - Cisco's software-defined solution for campus networks. - Provides automation, segmentation, and policy control. SD-WAN: - Software-defined wide area networking. - Uses centralized control to manage WAN connections. - Provides dynamic path selection, improved performance, and reduced costs.